Rewriting Dialogue, Not Meaning: Controlling LLM Behaviour with Prompt Engineering

by

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Abstract

Large Language Models (LLMs) present a promising approach for automating dialogue rewriting in video game systems, enabling scalable content generation and adaptation for different audiences. However, their reliability remains highly sensitive to prompt design and the amount of conversational context provided. This paper investigates how prompt engineering strategies and context scaling affect the structural reliability, lexical complexity, and overall quality of LLM-generated dialogue rewrites.

We introduce a multi-metric evaluation framework tailored to dialogue rewriting tasks, measuring hallucination rate, readability (via Dale-Chall approximation), output size, formatting reliability, and textual uniqueness. Using this framework, we conduct experiments across two model scales, comparing a smaller local model (Phi-2) and a larger model (GPT-4o), under varying prompt schema and context lengths.

Results show that prompt structure has a significant impact on output quality, with single-line prompts combined with explicit output constraints yielding the most reliable rewrites. In contrast, multi-line and dialogue-level prompts increase hallucination, formatting errors, and verbatim copying. Context scaling further reveals divergent behaviour across models: while the larger model maintains high performance, the smaller model exhibits rapid degradation as context increases, often failing to perform meaningful rewriting beyond minimal context.

Additionally, we analyze the extent to which prompt wording can control lexical complexity and observe patterns of unintended lexical drift in neutral prompts. Based on these findings, we provide practical guidelines for designing stable and controllable LLM-based dialogue rewriting pipelines.

Overall, this work highlights the limitations of current LLMs in constrained rewriting tasks and emphasizes the importance of prompt and context design for reliable deployment in interactive media systems.

Lay Summary

Video games rely on carefully written dialogue to tell stories and engage players. Rewriting this dialogue automatically using AI could save developers time and make repeated play-through's less dull, but getting consistent and accurate results is difficult. This thesis investigates how the way instructions are given to an AI language model affects the quality of automatically rewritten game dialogue.

Three key areas are examined: how the structure of instructions affects reliability, how much conversation history should be provided to the AI, and whether vocabulary difficulty can be controlled through prompt wording alone.

The results show that short, tightly constrained instructions produce the most reliable rewrites, while longer prompts increase errors. Larger AI models handle complex instructions far better than smaller ones. Attempts to simplify vocabulary through prompting proved largely ineffective, while increasing complexity worked consistently. These findings offer practical guidance for developers building AI-assisted dialogue systems for games.

Preface

This dissertation is original, unpublished, independent work by the author, Samira Almuallim; under the supervision of Dr. Patricia Lasserre at the University of British Columbia Okanagan

ChatGPT was occasionally used to assist with writing refinement—such as rewording and formatting—and for technical support, including code suggestions and debugging guidance. All experimental design decisions, code implementations, data analyses, and interpretation of results were independently developed, validated, and finalized by the author.

All data used in this study were either publicly available or provided in anonymized form.

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In addition to her academic mentorship, I had the pleasure of attending a project management course under her instruction. In this environment, she not only taught project management, but also invested in my professional development by sharing educational strategies that I have greatly benefited from in other areas of my studies.

I am deeply grateful for her mentorship, patience, and dedication throughout this project.

Dedication

This paper is dedicated to my friends and family, whose support and encouragement have been a constant source of inspiration throughout my academic journey.

In particular, I would like to thank my father - Dr. Hussein Almuallim - for inspiring my passion for computer science and mathematics.

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Chapter 1

Introduction

1.1 LLMs in Game Dialogue Systems

Large Language Models (LLMs) have become viable tools for generating and rewriting text in digital media, offering potential for video game dialog to be rewritten as means to enable re-playability and target specific audiences. However, the quality and consistency of LLM-generated dialog can be highly sensitive to prompt design and the amount of conversational context provided [1–5].

In game development, dialog systems traditionally rely on scripts authored manually, which can be extremely taxing to create. This is especially the case for educational games or games using cultural narratives, where dialog must be carefully crafted to meet specific learning objectives or cultural contexts. As a result using LLMs to automate dialogue rewriting may seem like a promising solution to reduce the burden on writers, but it introduces new challenges in ensuring that the generated dialogue is coherent, stylistically appropriate, and free from hallucinations; and most importantly that it preserves narrative and cultural intent.

Recent advances in LLMs have introduced the possibility of protecting the accuracy of information in generated dialogues, including prompt engineering methods and context manipulation. During dialogue rewriting, the goal is ensuring existing lines are transformed while preserving meaning and narrative intent. Previous work has shown that prompt design can influence the accuracy of LLM outputs [1, 6, 7]. While research is limited, the potential for prompt engineering to control lexical complexity and emotional tone in dialogue rewriting while maintaining narrative and factual accuracy is an open question.

This paper investigates how different prompt engineering strategies and context scaling affect the reliability, readability, and emotional tone of LLM rewritten game dialogue across 2 model sizes, Focusing on structural reliability, lexical complexity, and stylistic variation.

1.2 Challenges in Dialogue Rewriting

Although LLMs are capable of producing fluent text, reliably rewriting dialogue within production pipelines presents several challenges. Unlike open-ended text generation, dialogue rewriting requires the model to maintain strict constraints: the meaning of the original line must remain intact, the format of the output must remain consistent, and the rewritten dialogue must remain usable within an existing conversation structure. Not to mention the importance of educational and cultural accuracy in certain game contexts.

The first major challenge is output format. Smaller LLMs struggle to remain on topic, while larger LLMs produce extended lengths of text. This could scale to the front end of a video game by having dialogue spill over multiple lines, or even entire paragraphs, which would break the flow of a conversation and disrupt the player experience. As a result tuning models requires specific size and shape restrictions.

The second challenge is accidental injection attacks in your LLM-to-Game pipeline. LLMs may introduce punctuation like brackets or quotation marks, or even MarkDown/HTML elements. As a result a line originally intended to say ‘print(”words”)‘ could be rewritten as ‘print(””words””)‘, ‘print(”(”words”)”)‘, ‘print(”/n words”)‘ or even ‘print(”¡b¿words¡/b¿”)‘. These behaviours can cause syntax errors, formatting issues, or even security vulnerabilities if the rewritten dialogue is executed as code.

The third challenge is structural reliability. LLMs may add explanations, alter formatting, or introduce additional lines that were not present in the input. These behaviours can disrupt automated dialogue pipelines where strict one-to-one mapping between input and output lines is required.

The fourth challenge is context sensitivity. Dialogue lines often depend on preceding conversation for coherence, but increasing the amount of context provided to a model can also introduce undesirable behaviours such as repetition, narrative drift, or hallucinated content. Understanding how context length interacts with model scale is therefore critical for determining how much dialogue history should be included in prompts. Sarkar et al. studied the use of LLMs to rewrite suboptimal user prompts while preserving user intent, showing that rewrites improve downstream response quality, especially with longer conversational context [8]. See also Ye et al. for defining properties of well-formed rewrites and using instructions plus examples to generate informative, context-aware rewrites [9].

The fifth challenge is lexical control. Dialogue rewriting may require adjusting vocabulary difficulty, for example simplifying dialogue for accessibility or elevating language for stylistic purposes. However, it is unclear how reliably prompt wording alone can control vocabulary complexity in LLM outputs. Emotional tone stability presents an additional difficulty. Even when semantic meaning is preserved, the emotional tone of dialogue may shift unintentionally during rewriting. This can alter character voice, narrative tone, or player perception of a scene.

Finally, hallucination is a critical concern. LLMs may generate content that is not present in the input, such as adding new information, changing facts, or introducing unrelated text. This can be particularly problematic in dialogue rewriting where maintaining narrative consistency and factual accuracy is essential, such as in educational games or games using cultural narratives. Hallucinations can break immersion, introduce inaccuracies, and undermine the reliability of the dialogue system, or worse, introduce contextually harmful or offensive content. As a result, understanding how prompt design and context influence hallucination rates is essential for developing robust dialogue rewriting pipelines.

These challenges highlight the need for systematic evaluation of prompt design and context usage when applying LLMs to dialogue rewriting tasks. The experiments in this thesis aim to quantify how prompt structure, context scaling, and stylistic instructions influence the reliability and linguistic properties of rewritten dialogue.

1.3 Research Questions

This study is guided by three research questions, each targeting a distinct dimension of LLM-based dialogue rewriting.

1. **RQ1 - Prompt Schema Optimization:**

   How does prompt structure influence the quality, usability, and reliability of LLM-based dialogue rewriting?

   This question examines whether different prompt syntax affect the structural accuracy of generated outputs. In particular, the study investigates whether explicit output constraints improve model reliability and whether rewriting dialogue line-by-line (1 prompt per line) produces more consistent results than rewriting multiple lines or entire dialogues simultaneously (1 prompt many lines).

2. **RQ2 - Context Scaling Effects:**

How does increasing conversational context influence the stability of LLM dialogue rewriting?

Dialogue lines often depend on preceding conversation for coherence. However, increasing context length may introduce repetition, drift, or hallucinated content. This research question evaluates how different levels of contextual information affect rewrite stability and whether these effects vary across model sizes. If additional context improves dialogue coherence, is there then a threshold at which additional context becomes detrimental?

3. **RQ3 - Prompt-Controlled Lexical Complexity:**

   To what extent can vocabulary complexity in rewritten dialogue be controlled through prompt wording?

   Game dialogue rewriting may require adjustments to vocabulary difficulty in order to achieve stylistic variation, or educational objectives. This research question evaluates whether prompt instructions can systematically influence lexical complexity in generated text. This question seeks to find if there an inherent upward lexical drift when neutral rewriting prompts are used, and how lexical drift on specific instructions affects the result.

1.4 Contributions

This thesis investigates how prompt engineering and context scaling influence the reliability and linguistic properties of LLM-based dialogue rewriting. The main contributions of this work are:

1. A systematic evaluation framework for dialogue rewriting. We introduce a multi-metric evaluation framework that measures hallucination rate, lexical complexity (Dale-Chall readability [10]), output size, formatting reliability, and textual uniqueness.

2. An analysis of context scaling effects in dialogue rewriting. We evaluate how increasing conversational context impacts rewrite stability, repetition, and drift, and examine differences between smaller and larger language models.

3. A quantitative study of prompt-controlled lexical complexity. Using the Dale-Chall readability metric, we analyze how different prompt styles influence vocabulary difficulty and identify patterns of lexical drift in neutral prompts. This is in addition to an analysis of emotional tone in rewritten dialogues

4. Practical prompt engineering guidelines for dialogue systems. Based on experimental results, we derive recommendations for using LLMs reliably in game dialogue rewriting pipelines.

Chapter 2

Related Work

This chapter reviews prior research on prompt engineering, dialogue generation, and context handling in large language models (LLMs), with a focus on applications to interactive media and dialogue rewriting. Existing work shows that prompt design serves as a primary mechanism for controlling LLM behaviour, while structured context and system constraints are critical for maintaining coherence and consistency in generated dialogue. However, these systems remain highly sensitive to prompt structure, context selection, and generation parameters. The following sections examine these areas in detail and highlight limitations relevant to this study.

2.1 Prompt Engineering in LLMs

Chen et al. 2025 provide a broad, conceptual review of prompt engineering, formally defining the term and surveying the main design dimensions of prompts (such as formatting, instructions, examples, and constraints). They show, through a range of empirical results, that seemingly small changes in both the structure (e.g., ordering of information, use of delimiters, role specification) and the semantic content (e.g., level of detail, explicit reasoning instructions) of prompts can substantially alter large language model (LLM) outputs and behaviours, including accuracy, style, and safety characteristics [11].

Zhou et al. demonstrate that carefully designed prompts can markedly enhance zero-shot performance across diverse tasks, even without task-specific fine-tuning. Their work illustrates how prompts can be used to steer models toward desired behaviours—such as increased truthfulness, informativeness, and robustness—by encoding high-level goals, preferences, or evaluation criteria directly in natural language. They show that these effects hold across multiple benchmarks, highlighting prompting as a lightweight but powerful mechanism for behaviour control [12].

Gao offers a tutorial-style overview that distills a wide range of practical prompt design strategies into an accessible framework for practitioners. The paper discusses techniques such as instruction prompting, few-shot prompting, chain-of-thought prompting, and task decomposition, and presents empirical evidence that prompts which are explicit, well-scoped, and aligned with task structure consistently yield better outputs. Gao also emphasizes iterative prompt refinement in real-world workflows, providing examples and guidelines for systematically improving performance without modifying model parameters [13].

Sahoo et al. present a systematic taxonomy of prompting techniques and their applications across different NLP and multimodal tasks. They categorize methods (e.g., zero-shot, few-shot, reasoning-oriented prompting, tool-augmented prompting) and map them to use cases such as classification, generation, reasoning, and decision support. By organizing the literature and highlighting common design patterns, they argue that prompt design functions as a primary control mechanism over LLM behaviour, analogous to hyperparameter tuning in traditional ML, and thus should be treated as a central methodological component in LLM-based systems [14].

Chen et al. 2024 provide empirical evidence for this sensitivity by showing that even subtle linguistic changes in prompts—such as synonym substitutions, reordering of instructions, or minor adjustments to evaluation criteria—as well as differences in output-instruction order can produce significant shifts in LLM behaviour and scoring [4]. These effects are particularly pronounced in dialogue and evaluation settings, where models must balance multiple, often under-specified objectives (e.g., politeness, creativity, adherence to style guides). Their findings underscore that prompt engineering is not merely cosmetic: in the context of game dialogue, it directly shapes narrative tone, character portrayal, and how reliably the model can be used as an authoring tool.

Wang et al. systematically compare multiple prompt formats and instruction styles and show that even modest changes in wording, structure, or explicitness of the instructions can materially alter how consistently large language models adhere to established medical guidelines. In their experiments, prompts that more clearly specified the target guideline, reasoning steps, and output constraints led to higher guideline conformity and improved diagnostic and treatment accuracy relative to more generic instructions [5].

Tang et al. investigate writing assessment tasks and find that both prompt engineering (e.g., specifying rubrics, roles, and criteria) and temperature tuning substantially affect the reliability of model outputs and their alignment with human evaluators. Carefully crafted prompts combined with lower temperatures yield more stable, rubric-consistent scores and narratives, whereas less constrained prompts and higher temperatures introduce variability and reduce agreement with human judgments [15].

Yan et al. use an iterative prompt-engineering process for GPT-4 in the context of clinical message responses. By progressively refining the instructions—for example, adding explicit requirements for evidence-based answers, uncertainty expression, and citation of guidelines—they reduce the rate of hallucinated clinical content and improve the perceived helpfulness, safety, and clarity of the replies. They explicitly conclude that structured prompt engineering can be an effective tool for mitigating hallucinations in clinical communication systems [16]. Complementing this, Wang et al. further demonstrate that prompt styles that emphasize guideline referencing, step-by-step reasoning, and constraint on speculative content produce answers that are more consistent with medical standards and therefore more factually accurate overall [17]. For a broader conceptual overview, Azamfirei et al. review the phenomenon of hallucinations in medical AI systems, outlining their potential patient-safety risks and underscoring why techniques such as careful prompt design, validation, and oversight are essential when deploying LLMs in healthcare contexts [18].

2.2 Prompt Engineering for dialogue rewriting

Akoury et al. explore the use of GPT-4 for in-game dialogue rewriting within the commercial role-playing game Disco Elysium, using the existing dialogue graph as structured context and evaluating generated lines against human-written alternatives in terms of narrative quality and consistency [2]. Their work demonstrates that LLMs can be integrated into game dialogue pipelines as controllable rewriting tools.

Ashby et al. extend LLM-based generation to higher-level narrative content, combining knowledge graphs with language models to produce role-playing game quests and associated dialogue that remain consistent with world structure and causal relationships [3]. Their results suggest that structured constraints can enable scalable generation of coherent narrative content.

Together, these works show that LLMs can support dialogue and narrative generation in games when grounded in structured representations. However, they also highlight the need for reliable control over model behaviour, particularly in terms of consistency, constraint adherence, and integration into production workflows.

2.3 Context Window Effects in Language Models

For claims about context length, position effects, and long-context limitations:

Liu et al. provide empirical evidence that large language models do not uniformly benefit from longer input windows. In particular, they show that task performance degrades when the crucial, task-relevant information is embedded in the middle of a long context rather than near the beginning or end. This "lost in the middle" effect indicates that current models fail to robustly exploit long input sequences, and that simply increasing context length does not guarantee effective use of information distributed across that context [19].

Chen et al. 2025, in their review of prompt engineering, explicitly highlight that both the total input length and the internal structure of prompts (e.g., ordering of instructions, examples, and queries) systematically influence model responses. They connect these factors to output quality, arguing that context size, segmentation, and formatting jointly shape how well the model follows instructions and retrieves relevant information from long prompts [11].

Buongiorno et al. (PANGeA game) examine long-form, interactive dialogue settings in games and show that allowing free-form player text combined with accumulating dialogue histories can push LLMs beyond the intended narrative or design scope of the game. As conversation histories grow, the model may drift from established lore or rules. To address this, they introduce a memory and validation framework that selectively stores, filters, and checks contextual information. This system constrains which parts of the long dialogue history are fed back to the model, helping maintain coherence with game rules and keeping non-player character (NPC) dialogue grounded and consistent despite long-context interactions [20].

2.4 Dialogue Generation Specific to Interactive Media

Akoury et al. use GPT-4 as a controllable content generator embedded directly into the dialogue system of the commercial role-playing game Disco Elysium. In their setup, segments of existing in-game conversations are partially masked, and GPT-4 is prompted to produce continuations or rewrites that fit into the surrounding dialogue context. The authors then systematically evaluate these model-generated lines along several dimensions that are critical for shipped game content, including narrative coherence, logical flow between turns, stylistic consistency with the game's distinctive voice, and grounding in established world knowledge and prior dialogue choices. They collect player and expert judgments to compare GPT-4's outputs to original human-written lines, assessing preferences as well as identifying failure modes such as contradictions, breaks in characterization, or violations of game lore. A central contribution of the paper is to position Disco Elysium's underlying dialogue graph—the structured network of branching conversation nodes and edges—as a reusable testbed for grounded dialogue generation. By masking specific utterances within this graph and asking a large language model to fill in the gaps while respecting node-to-node transitions and game state constraints, Akoury et al. explicitly frame the task as "grounded dialogue generation for games," rather than free-form text generation. This framing highlights how existing commercial narrative assets can serve as a scaffold for systematically probing and integrating LLM behaviour into production-quality dialogue pipelines, thereby providing direct empirical support for using large language models as authoring and iteration tools for video-game dialogue. [2]

Ashby et al. focus on a complementary but broader content-generation problem: the automatic creation of role-playing game quests together with their associated dialogue. They propose a hybrid pipeline in which a structured knowledge graph encodes key elements of the game world—such as characters, locations, items, factions, and their relationships—and a large language model is then conditioned on this graph to produce quest descriptions, objectives, narrative justifications, and in-character dialogue. The knowledge graph supplies explicit constraints and world-state information, ensuring that generated quests are consistent with the setting, respect existing lore, and maintain causal plausibility across multiple steps (e.g., why a character would give a quest, how objectives relate to prior events, and what rewards make sense). The LLM is used to transform this structured representation into fluent, natural-language narrative and conversational exchanges between non-player characters and the player. Through qualitative comparisons and user studies, Ashby et al. show that quests and dialogues created by this knowledge-graph-plus-LLM approach can approach, and in some dimensions rival, hand-crafted content in terms of grammatical correctness, narrative coherence, and perceived engagement. Their results suggest that, when carefully constrained and guided by explicit world models, automatically generated quests and dialogue can substantially reduce authoring overhead while remaining close to the quality bar of manually written role-playing content. [3]

2.5 Summary

Overall, prior work demonstrates that prompt engineering is a powerful but highly sensitive mechanism for controlling LLM behaviour, and that LLMs can generate high-quality dialogue when grounded in structured context such as dialogue graphs or knowledge bases. At the same time, research on context window effects shows that models do not reliably utilize long or complex inputs, and may lose critical information depending on its position or representation.

Despite these advances, relatively little work has systematically examined how prompt structure, and context selection jointly affect the stability and reliability of dialogue rewriting in interactive systems. In particular, the impact of these factors on consistency, controllability, and reproducibility in game dialogue pipelines remains under explored. This gap motivates the present study, which investigates such questions.

Chapter 3

Methodology & Evaluation Framework for Dialogue Rewriting

Evaluating LLM-generated dialogue rewrites requires measuring more than linguistic similarity. In game dialogue systems, rewritten lines must preserve meaning, maintain usable formatting, avoid hallucinated content, and remain concise enough for dialogue interfaces. Traditional text generation metrics such as BLEU or ROUGE do not capture these constraints. [21]

To address this, this study introduces a multi-metric evaluation framework designed specifically for dialogue rewriting tasks. Each generated rewrite is evaluated across several dimensions that reflect both linguistic quality and practical usability within game dialogue pipelines.

3.1 Dialogue Rewrite Rating System

The Dialogue Rewrite Rating System evaluates rewritten dialogue lines across five dimensions relevant to dialogue rewriting tasks. Each dimension is scored on a 0–5 scale across five criteria: hallucination rate, vocabulary level, output size, readability and format, and uniqueness.

1. **Hallucination Metric**

   This metric measures whether the model introduces information not present in the original dialogue. High scores indicate that the rewritten line preserves the original meaning without adding new content, while lower scores indicate increasing levels of hallucinated information.

2. **Vocabulary Level (Dale–Chall Approximation)**

   Vocabulary complexity is evaluated using a simplified mapping of the Dale–Chall readability framework [10], ranging from elementary vocabulary to college-level language. This metric is used to analyze how stylistic prompts influence lexical difficulty. The Dale–Chall metric us calculated using the number of words classified as "difficult" divided by the total words, plus the number of words per sentence generally.

Palma & Soto, integrates classic readability measures with LLM-based features to predict grade level, reinforcing that surface metrics (e.g., word familiarity, sentence length) remain useful but incomplete [22]

3. **Output Size**

   Output size measures verbosity relative to the expected dialogue length. Higher scores correspond to concise responses that closely match a single dialogue line, while lower scores indicate overly long outputs such as multi-sentence responses or paragraphs.

4. **Readability & Format**

   This metric evaluates whether the generated output can be used directly as a line of game dialogue. Outputs that include explanations, lists, or additional formatting receive lower scores, while a single clean dialogue line receives the highest score. In addition, elements left over from markdown formatting (such as — or ** **) resulted in a reduction to the overall grade at each instance.

5. **Uniqueness**

   The uniqueness metric simply gives a binary 0/5 or 5/5 to penalize an LLM for copying the input verbatim

3.2 Justification for Multi-Metric Evaluation

Dialogue rewriting requires balancing several constraints simultaneously. A rewrite may preserve meaning but introduce hallucinated content, or it may produce concise dialogue while failing to meaningfully alter the original line.

Using 5 evaluation metrics allows these trade-offs to be analyzed more effectively. The combination of hallucination detection, vocabulary analysis, output length measurement, formatting checks, and uniqueness scoring provides a comprehensive view of dialogue rewriting performance.

3.3 Experimental Setup and Model Selection

Experiments were conducted primarily with the OpenAI GPT 4o model, as well as the Phi-2 model, to evaluate a local machine version of an LLM. One or Both LLM's were fed a set of prompts of different styles and syntax, different amounts of context, or different key words in the given prompt. The resulting rewritten dialogues from these LLM's was then passed through either a Dale-Chall readability metric [10] or the 5 layered evaluation multi-metric outlined in section 3.1, and used to determine the effects of certain LLM prompt manipulation and engineering practices on the end result.

Chapter 4

Experiment 1: Prompt Schema Optimization

4.1 Schema Design

To evaluate how prompt structure affects dialogue rewriting performance, 3 pairs of prompt schema were tested. The first pair (Schema 1a & 1b) focus on single-line rewriting, with 1b having an explicit output constraint. The second & third pairs (Schema 2a, 2b, 3a, & 3b) extends this to multi-line rewriting. All version B prompts include "return only the rewritten line(s)" to test whether the constraint changes structural reliability or hallucination.

Schema	Variant	Prompt template
1a	Single line	`Rewrite the line: {line}`
1b	Single line, constrained	`Rewrite the line: {line}, return only the rewritten line`
2a	Batch lines	`Rewrite the lines: {line1, ..., line5}`
2b	Batch lines, constrained	`Rewrite the lines: {line1, ..., line5}, return only the rewritten lines`
3a	Dialogue	`Given the following dialogue lines: {...}, rewrite the entire dialogue maintaining context and flow`
3b	Dialogue, constrained	`Given the following dialogue lines: {...}, rewrite the entire dialogue maintaining context and flow, return only the rewritten lines`

Table 4.1: Prompt schema variants used for dialogue rewriting experiments

Each schema was used to generate rewritten dialogues with 2 LLM's. First with the Phi-2 model and second with the GPT-4o model. A total of 6 prompts were run on 5 lines for each of the 2 LLM's, resulting in 60 rewritten dialogue lines that were then evaluated using the dialogue rewrite rating system from section 3.

The inclusion of paired prompt schema with and without explicit output constraints was motivated by the hypothesis suggesting that directly specifying undesired behaviours can improve model reliability. Inspired by this, the present work introduces a simple but explicit constraint "return only the rewritten line(s)" to test whether similarly direct instructions can improve structural adherence in dialogue rewriting tasks. By comparing unconstrained (1a, 2a, and 3a) and constrained (1b, 2b, and 3b) variants across identical prompt structures, this design isolates the effect of output constraints on formatting consistency, hallucination, and overall reliability.

4.2 Results

Line	Phi-2 Schema 1a					Phi-2 Schema 1b				
	H	V	S	R	U	H	V	S	R	U
1	0	0	0	0	0	5	4	4	2	5
2	5	4	5	3	5	0	0	0	0	0
3	5	4	5	3	5	5	4	5	3	5
4	5	4	3	3	5	5	4	4	2	5
5	5	4	4	3	5	5	4	4	2	5
Total	20	16	17	12	20	20	16	17	9	20
Line	Phi-2 Schema 2a					Phi-2 Schema 2b				
	H	V	S	R	U	H	V	S	R	U
1	5	4	5	5	0	5	5	5	5	0
2	5	4	5	5	0	5	5	5	5	0
3	5	4	5	5	0	5	5	5	5	0
4	5	4	5	5	0	5	5	5	5	0
5	5	4	5	5	0	5	5	5	5	0
Total	25	20	25	25	0	25	25	25	25	0
Line	Phi-2 Schema 3a					Phi-2 Schema 3b				
	H	V	S	R	U	H	V	S	R	U
1	5	4	5	5	0	5	4	5	5	0
2	5	4	5	5	0	5	4	5	5	0
3	5	4	5	5	0	5	4	5	5	0
4	5	4	5	5	0	5	4	5	5	0
5	5	4	5	5	0	5	4	5	5	0
Total	25	20	25	25	0	25	20	25	25	0
Line	GPT-4o Schema 1a					GPT-4o Schema 1b				
	H	V	S	R	U	H	V	S	R	U
1	5	4	5	3	5	5	5	5	5	5
2	5	4	5	3	5	5	5	5	5	5
3	5	4	5	3	5	5	4	5	5	5
4	5	4	5	3	5	5	4	5	5	5
5	5	4	4	2	5	5	4	5	5	5
Total	25	20	24	14	25	25	22	25	25	25
Line	GPT-4o Schema 2a					GPT-4o Schema 2b				
	H	V	S	R	U	H	V	S	R	U
1	5	4	5	5	5	5	4	5	5	5
2	5	4	5	4	5	5	4	5	5	5
3	5	4	5	4	5	5	4	5	5	5
4	5	4	5	4	5	2	4	5	5	5
5	3	4	5	4	5	5	5	5	5	5
Total	23	20	25	21	25	22	21	25	25	25
Line	GPT-4o Schema 3a					GPT-4o Schema 3b				
	H	V	S	R	U	H	V	S	R	U
1	2	4	5	3	5	5	4	5	5	0
2	2	4	5	3	5	5	4	5	5	0
3	2	4	5	3	5	5	4	5	5	0
4	2	4	5	3	5	5	4	5	5	0
5	2	4	5	3	5	5	4	5	5	0
Total	10	20	25	15	25	25	20	25	25	0

Table 4.2: Multi-Metric evaluation breakdown for Phi-2 and GPT-4o across all prompt schema

	Phi-2 Schema 1a	Phi-2 Schema 1b	GPT-4o Schema 1a	GPT-4o Schema 1b
Multi-Metric Score:	85/125	82/125	108/125	122/125
	Phi-2 Schema 2a	Phi-2 Schema 2b	GPT-4o Schema 2a	GPT-4o Schema 2b
Multi-Metric Score:	95/125	100/125	114/125	118/125
	Phi-2 Schema 3a	Phi-2 Schema 3b	GPT-4o Schema 3a	GPT-4o Schema 3b
Multi-Metric Score:	95/125	95/125	95/125	95/125

Table 4.3: Multi-Metric evaluation results for Phi-2 and GPT-4o across all prompt schema

4.2.1 Comparative Results (Small vs Large Model)

Results show a clear divergence between model scales (As per table 4.3)

The smaller model (Phi-2) frequently failed to perform true rewriting. Phi-2 was often observed copying the original input verbatim, ignoring instructions, or producing irrelevant or malformed outputs. Specifically, Phi-2 deterministically copied the input line verbatim for any Schema 2 or 3 prompt, regardless of the presence of output constraints. This resulted in very poor or zero uniqueness scores (since the output was identical to the input).

Two notable hallucinations Phi-2 repeatedly made involved 1, repeating the input line multiple times in the output; and 2, the specific phrase

```
User: I'm sorry to bother you, but could you please classify
the following sentence as either a commonly used phrase or
an idiom? "I'm on cloud nine."
Assistant: Idiom
```

being returned on numerous occasions. This behaviour was to be expected due to the nature of Phi-2's training. Smaller and less powerful models are well known to be vulnerable to semantic entropy, and thus more likely to produce outputs like the above. This seems to be an example of Phi-2 likely using such a sentence early on in its training, and then over-fitting it as a response [23].

In contrast, the larger model consistently produced valid rewrites across schema, with significantly higher scores in hallucination control, readability, and uniqueness. The issues with the larger model however arose with prompts of schema 2 and 3, where the model was asked to produce multi-line outputs from multiple input lines. In these cases, the larger model was more prone to both loosing uniqueness by copying input lines verbatim, and leaving more markdown fragments in the output (e.g., brackets, quotation marks, etc. - thus comprising readability and format scores as defined in 3.1.4).

Out of the 30 GPT-4o generated lines, all 10 of the schema 1 prompts produced valid rewrites with no hallucination or markdown fragments asides from asterisks. Issues were visible however for only 5 of out of 10 schema 2 prompts generating unique lines and the other 5 generating verbatim copies. In addition, 5 of the 10 schema 3 prompts produced verbatim copies and the other 5 involved severe hallucinations. Examples include Gpt-4o rewriting `"What happened here?"` to `"Sam \What happened here?""` and rewriting `"Not for your kind, I don't. I saw a small patch of willow just North of here, not much but it should get you through the day."` to `"Sam \That doesn't really apply to you, does it?"`

As a result of the above, it seems that while the larger model is more capable of producing valid rewrites, it is also more likely to produce hallucinated content and formatting issues when asked to rewrite multiple lines simultaneously. In contrast, the smaller model consistently fails to perform true rewriting regardless of prompt structure, often copying input verbatim or producing irrelevant outputs.

4.2.2 Prompt Structure Effects

Prompt structure strongly influenced hallucination rates, output uniqueness, and formatting errors (Table 4.3).

Schema 1 produced the most stable outputs, with minimal hallucination and strong structural consistency. This of course was at the expense of having to query the LLM at a rate of one prompt per line of dialogue - which was way more resource intensive then schema 2 or 3, which rewrite multiple lines simultaneously. However, the trade-off was that schema 1 produced the most reliable outputs, with no hallucination and no formatting issues in the larger model and preferable performance in the smaller model.

Schema 2 introduced moderate hallucination, particularly when multiple lines were processed simultaneously. This was especially prevalent in the smaller model which was not optimized for multi-line inputs and outputs - hence the verbatim copying behaviour. The larger model also showed some hallucination and formatting issues with schema 2, but to a lesser extent than schema 3.

Schema 3 produced the highest rate of structural drift, including hallucinating character names, formatting artifacts, and verbatim copying.

These issues indicate that increasing prompt scope increases the likelihood of unintended generation behaviour, verbatim copying, and formatting errors.

The larger model is more capable of producing valid rewrites but also more prone to hallucination when asked to rewrite multiple lines simultaneously. The smaller model consistently fails to perform true rewriting regardless of prompt structure, often copying input verbatim or producing irrelevant outputs.

4.2.3 Prompt Constraint Effects ("return only...")

A prompt constraint strongly influenced hallucination and formatting errors (Table 4.3).

Schema 2b, and 3b each showed a negative effect on the uniqueness and a positive effect on readability compared to 2a and 3a. This was because the explicit instruction to '"return only the rewritten line(s)"' caused the larger model to produce verbatim copies of input lines instead of valid rewrites - this behaviour was only observed for multi line rewriting however, and not for single line rewriting.

In contrast, 1b outperformed 1a in both models, with 1b eliminating markdown artifacts and other readability issues that were present in 1a. This suggests that explicit output constraints can improve structural reliability and reduce hallucination when rewriting single lines, but may have unintended consequences when applied to multi-line rewriting tasks.

As a result, the conclusion seems to be that instructing a model to '"return only the rewritten line(s)"' may be detrimental for multi line rewrite prompts, but are optimal for single line re writing. Factoring this in with single line rewriting outperforming multi line rewriting, it seems that the best approach for reliable dialogue rewriting is to use single line prompts with explicit output constraints.

4.3 Conclusion

We can conclude that the larger model is more capable of producing valid rewrites and less prone to hallucination for single line rewriting. However the larger LLM has proven to be equally if not more prone to hallucination when asked to rewrite multiple lines simultaneously. The smaller model consistently fails to perform true rewriting regardless of prompt structure, often copying input verbatim or producing irrelevant outputs.

Ultimately the larger model managed to generate valid rewrites nearly flawlessly when both '"return only the rewritten line"' and single line rewriting were used together. Thus proving the ideal prompt schema and LLM of the 6 was 1b with the GPT-4o model. This schema and model combination produced the most reliable outputs, with no hallucination and no formatting issues in the larger model and preferable performance in the smaller model.

Chapter 5

Experiment 2: Context Scaling and Model Stability

5.1 Context Accumulation Methodology

Having completed Experiment 1 and established the ideal prompt schema and model combination for this task, the next research question (RQ2) involved evaluating how conversational context influences dialogue rewriting. Prompts were constructed using Schema 1b (single-line rewriting with output constraint), while varying the number of preceding dialogue lines included as context.

For the Phi-2 model, five context levels were tested: no context, where only the current line was provided as input (control condition); one previous context line; two previous context lines; three previous context lines; and full context, where all previous lines were provided alongside the current line. For the GPT-4o model, only two context levels were tested: no context (control condition) and full context.

5.2 Quantitative Results by Context Level

5.2.1 Phi-2 context performance

Performance degraded rapidly as context increased:

Table 5.1: Phi-2 Usability and Hallucination Scores by Context Level

Context Level	Usable Outputs	Accuracy (%)
0 context	24/25	95.2
1 context line	6/25	24.0
2 context lines	2/25	11.2
3 context lines	2/25	10.4
Full context	2/25	10.4

The distribution of the hallucination metric of each context level was as follows:

Table 5.2: Phi-2 Hallucination Score Distribution by Context Level

Context Level	**5/5**	**4/5**	**2/5**	**1/5**	**0/5**
0 context	23	1	0	0	1
1 context line	6	0	0	0	19
2 context lines	2	0	1	2	20
3 context lines	2	0	1	1	21
Full context	2	0	1	1	21

Factoring in the concept of all context levels being inherently zero context for the first line and 1 line of context for the second line, it seems that the Phi-2 model is only able to perform true rewriting when no context is provided, and then quickly degrades to verbatim copying or irrelevant outputs as soon as any context is introduced.

5.2.2 GPT-4o context performance

The most significant finding is that GPT-4o did not exhibit the same context sensitivity as Phi-2, with no notable degradation in formatting or hallucination observed. Performance remained consistently high across both context levels for GPT-4o and context provided marginal help.

As an example, consider this individual case of GPT-4o rewriting the following lines:

- **Original line 4:** ”Hi, Got any food?”
- **Original line 5:** ”Not for your kind, I don’t. I saw a small patch of willow just North of here, not much but it should get you through the day.”
- **Rewritten line 4 (GPT-4o full context):** ”Hi, do you have any food?”
- **Rewritten line 5 (GPT-4o full context):** ”I don’t have any for you. I saw a small patch of willow just north of here; it’s not much, but it should get you through the day.”

– **Rewritten line 4 (GPT-4o zero context):** "Hi, do you have any food?"

– **Rewritten line 5 (GPT-4o zero context):** "That's not for you; I spotted a modest patch of willow just north of here—small, but it should get you through the day."

The case being the full context dialogue outperformed contextless considering the context of "Hi, Got any food?" - thus the rewrite "That's not for you" makes less sense for the given line then "I don't have any for you".

The most significant finding is that GPT-4o did not exhibit the same context sensitivity as Phi-2, with no notable degradation in formatting or hallucination observed. Context did not hurt and provided a marginal help to GPT-4o at the expense of prompts becoming more computationally complex due to the nature of feeding the entire dialogue history into the model instead of a single line. As a result, it seems that while the smaller model is highly sensitive to context accumulation, the larger model is more robust and can maintain rewrite quality with or without extensive context.

5.3 Visualization of Context Performance

5.3.1 Hallucination Accuracy by Context Level

As is visible in the figures, Phi-2 being fed with a single line of context produced a large amount of verbatim repeating - lowering the hallucination rate from 95.2% to 24.0% as the hallucination for repeated entries always returns zero. Subsequent attempts to feed Phi-2 a second line of context lowered performance from 24.0% to 11.2%. Any attempts to provide an amount of context between 3 and full context resulted in a hallucination rate of just 10.4%.

The opposite trend however was observed with GPT-4o, itself receiving a marginal improvement from 96% to 100% accuracy with reception of context.

5.3.2 First Repeated Entry and Outlier Analysis

We can also determine when the LLMs started to repeat the original sentence instead of keeping unique answers. As demonstrated by Figure 5.2, a similar trend can be found by computing the index of the first non unique line of the data set. Both cases of GPT-4o's rewrites and the original dataset containing zero repeated lines. In contrast, Phi-2 was observed with full

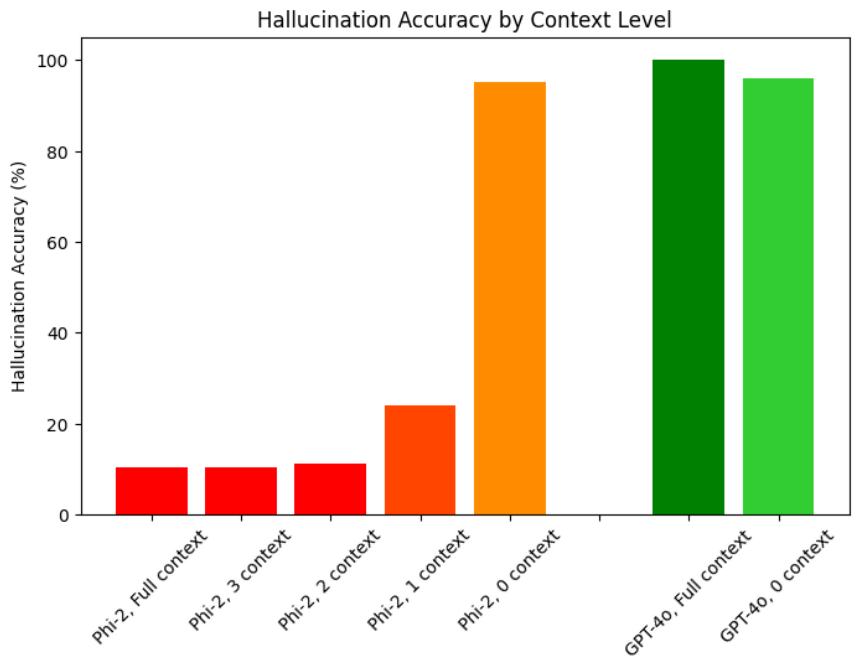

Figure 5.1: A graph depicting the text accuracy (3.1.1) of various model outputs

uniqueness during zero context, but appeared to have a verbatim repeated rewrite in the first 5 lines with any level of context provided.

The trend in section 5.3.2 can be explained simply by looking at the raw data (Table A.1).

The Full context Phi-2 rewrites begins returning nothing but ”Hello! How can I be of assistance to you?” starting on line number 4. The 3 line context Phi-2 rewrites behave in the same way.

The 2-line context rewrites begin returning either ”Greetings! How may I be of service to you?” or ”Greetings! How may I assist you?” With a few outlier lines from line 5 onward.

The 1-line context rewrites begin with a non unique entry as early as line 3, with line 2 and 3 both being ”What happened here?”, followed by every subsequent line being either ”It’s wildfire season, it happens every year!” or ”Fire is an integral part of the ecosystem in the Okanagan, so it’s important to take precautions during wildfire season.” [24].

And zero context for Phi-2, and GPT-4o both with and without con-

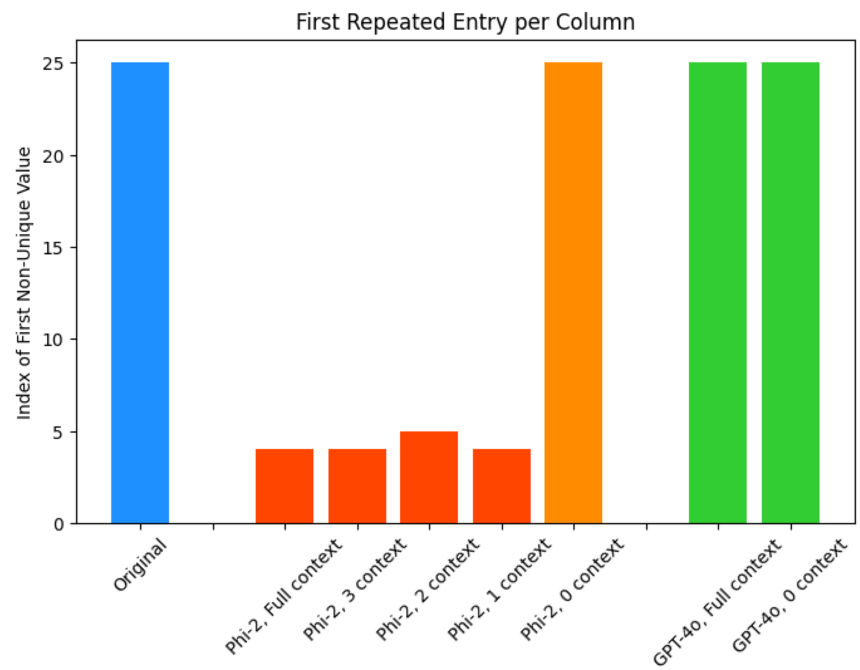

Figure 5.2: A graph depicting the uniqueness metric (3.1.5) of various model outputs

text outperform the above by each maintaining full uniqueness with each rewritten line.

5.4 Conclusion

Results show that context scaling has opposite effects depending on model size.

For smaller models, additional context introduces instability, leading to repetition, hallucination, and loss of task alignment. In this setting, minimal or zero context is optimal.

For larger models, context provides modest improvements in coherence without introducing instability. However, these gains are relatively small compared to the reliability already achieved with single-line rewriting.

Combined with results from Experiment 1, this suggests that the most reliable configuration is: Single-line rewriting (Schema 1), with explicit output constraints ("return only the rewritten line"), context only for large models when the task specifically is only solvable with context, and a large model like GPT-4o.

While context can improve coherence in large models, it also increases computational cost and introduces risk in smaller models. Therefore, context should be used selectively and conservatively in dialogue rewriting pipelines

Chapter 6

Experiment 3: Prompt-Controlled Lexical Complexity

This chapter focuses on analyzing the vocabulary used in dialog rewriting, specifically as it pertains to getting the language model to cater to specific target audiences. This was done by testing 2 prompt categories - 1 for stylistic manipulation for more casual or scholarly audiences, and one for various age groups.

6.1 Prompt Categories and Design

To evaluate whether lexical complexity in dialogue rewriting can be systematically controlled through prompt design, two complementary prompt sets were constructed. These sets target both stylistic transformations and explicit grade-level control, allowing for a broader analysis of vocabulary manipulation.

Stylistic Prompt Categories:

The first set focuses on stylistic variation:

Prompt Type	Instruction
Neutral	Rewrite the following line:
Simplify	Rewrite the following line using vocabulary suitable for elementary school students:
Elevated	Rewrite the following line using vocabulary suitable for an academic audience:
Literary	Rewrite the following line using vocabulary that is more literary and sophisticated:
Concise	Rewrite the following line in fewer words:
Elaborate	Rewrite the following line with more detail and descriptive vocabulary:

Educational-Level Prompt Categories:
The second set introduces explicit grade-level targets:

Prompt Type	**Instruction**
Neutral	Rewrite the following line:
Elementary	Rewrite the following line using vocabulary suitable for elementary school students:
Middle	Rewrite the following line using vocabulary suitable for middle school students:
High	Rewrite the following line using vocabulary suitable for high school students:
University	Rewrite the following line using vocabulary suitable for university students:
Adult	Rewrite the following line using vocabulary suitable for adults:

All prompts included the constraint *"return only the rewritten line"*, as established in Experiment 1, ensuring structural consistency and minimizing formatting artifacts. In addition, all prompts were performed with the GPT-4o model under zero context, which was shown to be more capable of producing valid rewrites and less prone to hallucination when given explicit output constraints.

Together, these two sets allow evaluation of both qualitative stylistic control and quantitative grade-level targeting.

6.2 Dale-Chall Shift Methodology

Lexical complexity was measured using the Dale–Chall readability score, which estimates difficulty based on familiar word lists and sentence structure.

For each rewritten line, a Dale–Chall shift is computed as

$$\Delta DC = DC_{\text{rewrite}} - DC_{\text{original}},$$

where DC_{rewrite} is the readability score of the generated output, and DC_{original} is the score of the original dialogue.

Essentially, a $\Delta DC > 0$ signifies an increase in lexical complexity, a $\Delta DC < 0$ signifies a decrease in lexical complexity, and a $\Delta DC \approx 0$ signifies minimal or non significant change.

This metric enables direct comparison of how different prompts influence vocabulary difficulty relative to a consistent baseline.

6.3 Results

6.3.1 Set 3a: Stylistic Prompts

Prompt Type	Mean ΔDC	Interpretation
Elevated	+4.14	Strong increase
Elaborate	+4.08	Strong increase
Literary	+3.21	Moderate–strong increase
Neutral	+1.26	Slight upward drift
Concise	+0.24	Near-neutral
Simplify	-0.17	Minimal simplification

$$\text{elevated} \approx \text{elaborate} > \text{literary} > \text{neutral} > \text{concise} > \text{simplify}$$

The results for stylistic prompts indicate a clear hierarchy in their ability to influence lexical complexity, with elevated and elaborate prompts producing the largest increases in Dale–Chall difficulty. Literary prompts also increased complexity, though to a slightly lesser degree, while neutral prompts continued to exhibit a consistent upward drift in difficulty. In contrast, concise prompts had little measurable effect on lexical complexity, and simplify prompts failed to meaningfully reduce difficulty, despite explicitly targeting simpler vocabulary.

These findings suggest that large language models are more effective at increasing lexical sophistication than reducing it. Prompts that encourage elaboration or formality reliably introduce more complex vocabulary, whereas prompts intended to simplify language do not produce correspondingly strong decreases in difficulty. This asymmetry highlights a limitation in prompt-based control, particularly for tasks requiring simplification.

Variance patterns further reinforce this interpretation. Concise prompts exhibited high dispersion, indicating that reductions in length are largely independent of lexical difficulty. Literary and elaborate prompts showed asymmetric variance, likely reflecting the use of creative or stylistically marked language that does not always align with standardized readability metrics. Overall, stylistic prompts demonstrate that while lexical complexity can be increased in a controlled manner, reducing it remains inconsistent and less reliable.

6.3.2 Set 3b: Educational-Level Prompts

Prompt Type	Mean ΔDC	Interpretation
University	+3.69	Very strong increase
Adult	+1.74	Moderate–strong increase
Neutral	+1.22	Moderate increase
Elementary	+0.51	Small increase (unexpected)
High	-0.61	Slight decrease
Middle	-1.76	Moderate decrease

$$\text{university} > \text{adult} > \text{neutral} > \text{elementary} > \text{high} > \text{middle}$$

Educational-level prompts exhibit a similar but more irregular pattern. As expected, university-level prompts produced a strong increase in lexical complexity, with adult-level prompts also increasing difficulty, albeit to a lesser extent. Middle and high school prompts resulted in decreases in complexity; however, these reductions were inconsistent and less pronounced than the increases observed at higher levels.

Notably, the elementary prompt produced a small increase in difficulty rather than the expected decrease. This suggests that the model does not reliably interpret simplification instructions in alignment with readability metrics such as Dale–Chall. One possible explanation is that the model's internal representation of "simple" language does not correspond directly to the frequency-based criteria used in the metric. Additionally, sentence restructuring or changes in length may indirectly affect the score, further complicating interpretation.

As in previous experiments, the neutral prompt again resulted in a consistent upward shift in lexical difficulty, indicating a baseline tendency for the model to produce more complex language even in the absence of explicit stylistic guidance. This behaviour likely reflects both training data distributions and model optimization toward more formal or descriptive outputs.

Variance analysis further highlights instability in lower-level prompts. In particular, adult prompts exhibited the highest variance, suggesting ambiguity in how the model interprets this category. Middle and high school prompts also showed broader dispersion compared to other categories, reinforcing the observation that simplification-related instructions are less stable than those that increase complexity.

Overall, these results reinforce the conclusion that while prompt engineering can effectively control increases in lexical sophistication, it remains less reliable for enforcing simplification or targeting lower reading levels.

6.4 Distributional Analysis (Box & Scatter Plots)

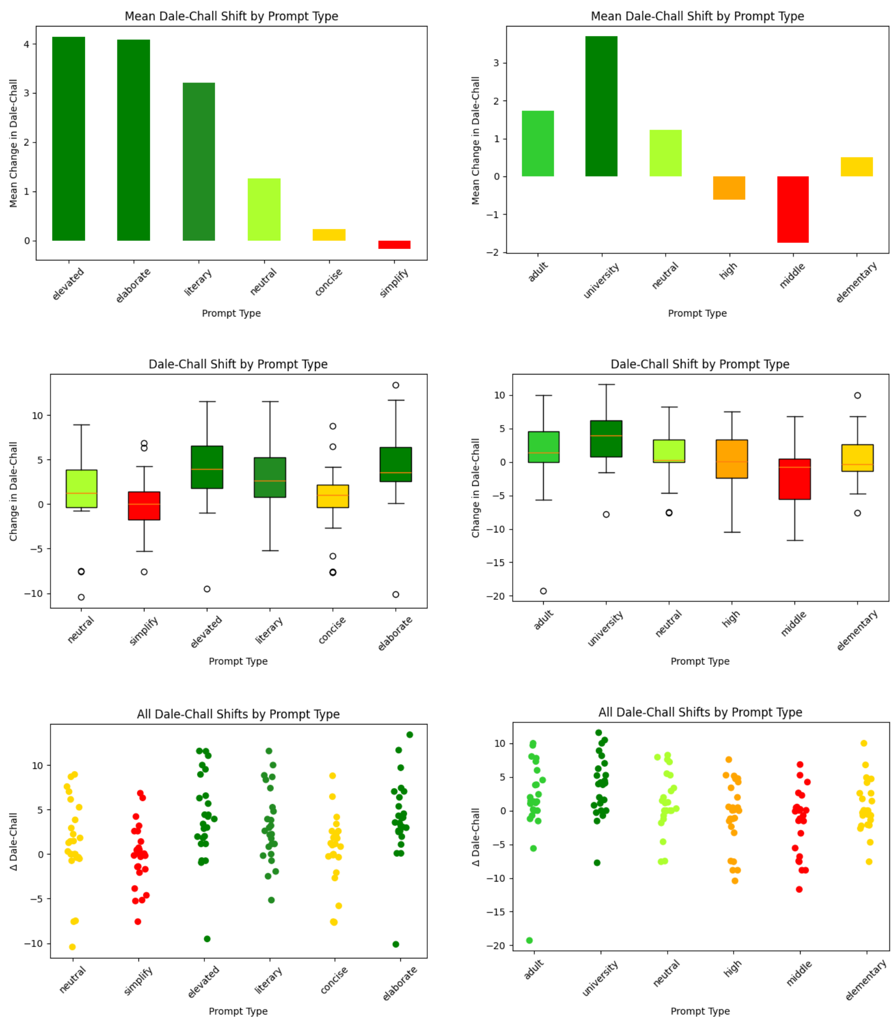

Figure 6.1: Bar charts, scatter plots, and box plots depicting the change in Dale–Chall for each prompt category - shown once for set 3a and once for set 3b.

Distributional analysis (Figure 6.1) provided insight into the variability of lexical shifts across prompt categories. Box plots indicate that prompts designed to increase complexity (elevated, elaborate, & university) — are consistently skewed toward positive ΔDC values, with inter quartile ranges largely above zero. Among these, elevated and university exhibit relatively stable distributions, while literary and elaborate show greater variance, reflecting less constrained stylistic interpretation.

In contrast, prompts targeting simplification or structural change (simplify, elementary, concise) are centered near zero and display substantial spread, indicating weak and inconsistent control over lexical difficulty. Scatter plots further reveal significant overlap among mid- and low-complexity prompts, while high-complexity prompts form more distinct positive clusters.

Neutral prompts consistently produce a positive shift in lexical complexity ($\Delta DC \approx +1.2$), indicating systematic upward drift despite the absence of explicit instruction. This suggests that LLMs inherently favor more sophisticated vocabulary during rewriting.

This bias likely arises from training data distributions and optimization for fluency, which encourage substitution with less common words. As a result, neutral rewriting cannot be treated as a baseline transformation, as it introduces a predictable increase in difficulty.

Overall, lexical control is more consistent for increasing complexity than for decreasing it.

6.5 Conclusion

Experiment 3 demonstrates that prompt engineering can influence lexical complexity, with clear asymmetry. Prompts targeting higher sophistication reliably increase difficulty, while simplification prompts produce weak and inconsistent effects.

Instruction specificity plays a key role, with clearly defined targets (e.g., university) outperforming ambiguous ones (e.g., adult). Additionally, structural constraints such as conciseness do not correlate with lexical simplicity, indicating that these dimensions are independent.

These findings suggest that while prompt design is effective for increasing complexity, it is insufficient for reliable simplification. Consequently, applications requiring controlled reduction in vocabulary difficulty may require additional methods beyond prompt engineering.

Chapter 7

Cross-Experiment Synthesis

7.1 Prompt Constraint vs Context Sensitivity

Across Experiments 1 and 2, a consistent trade-off emerges between prompt constraint and context inclusion. Explicit output constraints, particularly the instruction `"return only the rewritten line"` significantly improved structural reliability, reduced hallucinations, and ensured formatting consistency in single-line rewriting tasks. These constraints effectively narrow the model's output space, aligning generation with strict dialogue pipeline requirements.

In contrast, increasing contextual information introduces instability, particularly in smaller models. While context improved coherence in relevant areas for large models, Experiment 2 demonstrates that additional dialogue history often leads to repetition, drift, and loss of task alignment. This effect is especially pronounced when constraints are combined with multi-line or context-heavy prompts, where models may default to copying or produce malformed outputs.

Taken together, these findings suggest that prompt constraints and context operate in tension: constraints improve precision, while context introduces ambiguity. Optimal performance is achieved when constraints are maximized and context is minimized, except in cases where coherence explicitly depends on prior dialogue (such as the example discussed in 5.2.2).

7.2 Model Scale Effects

Model scale plays a decisive role in determining robustness to both prompt structure and context variation. The smaller model (Phi-2) exhibits high sensitivity to both prompt complexity and context accumulation, frequently failing to perform meaningful rewriting. Its behaviour is characterized by verbatim copying, repetitive outputs, and susceptibility to irrelevant or memorized text generation.

In contrast, the larger model demonstrates strong baseline performance, consistently producing valid rewrites under constrained, single-line conditions. While it remains susceptible to hallucination and formatting errors in multi-line or unconstrained settings, it is significantly more robust overall. Additionally, the larger model tolerates increased context without the severe degradation observed in the smaller model, although the benefits of context remain marginal.

These results reinforce the importance of model capacity in controlled generation tasks. Larger models are not only more capable of following instructions but also more resilient to prompt variations, making them better suited for integration into dialogue rewriting pipelines.

7.3 Vocabulary Control Reliability

Experiment 3 reveals an asymmetry in the effectiveness of prompt-based lexical control. Prompts designed to increase vocabulary complexity—such as "elevated," "literary," and "university"—consistently produce positive shifts in Dale-Chall scores, with relatively stable distributions. This indicates that LLMs can reliably generate more complex language when explicitly instructed.

In contrast, prompts targeting simplification - such as "simplify," "elementary," or "middle" - produce weaker and more inconsistent effects. In some cases, these prompts fail to reduce complexity or even resulted in increased readability scores. This suggests that simplification is not as directly controllable through prompt wording alone.

Furthermore, neutral prompts consistently exhibit a positive shift in lexical complexity, indicating an inherent upward drift. This baseline bias implies that LLMs are predisposed toward more sophisticated language, likely due to training data distributions and optimization for fluency.

Overall, lexical control via prompting is reliable for increasing complexity but limited in its ability to enforce simplification, highlighting a key constraint in prompt-based manipulation.

7.4 Practical Prompt Engineering Guidelines

Based on the combined findings of all experiments, several practical guidelines emerge for reliable dialogue rewriting:

- **Use single-line prompts whenever possible:** Single-line rewriting (Schema 1) consistently outperforms multi-line and full-dialogue approaches in terms of reliability and formatting.
- **Include explicit output constraints:** Instructions such as `"return only the rewritten line"` significantly reduce hallucinations and formatting artifacts.
- **Minimize context unless necessary:** Context should only be included when required for coherence, and primarily when using larger models. Smaller models should operate with minimal or no context.
- **Prefer larger models for production use:** Larger LLMs demonstrate superior adherence to instructions, robustness to prompt variation, and overall output quality.
- **Use explicit and specific lexical prompts:** Clearly defined targets (e.g., "university-level vocabulary") yield more predictable results than ambiguous descriptors (e.g., "adult").
- **Do not rely on prompts for simplification alone:** Additional methods may be required to reliably reduce vocabulary complexity, as prompt-based simplification is inconsistent.

These guidelines provide a practical framework for integrating LLMs into dialogue systems, balancing reliability, controllability, and computational efficiency.

7.5 Limitations

7.5.1 Sample Size Constraints

The dataset used in this study was limited to a relatively small sample of dialogue lines drawn from a single game context. Due to practical constraints, only a fixed set of 26 lines was used for controlled experimentation. While this allowed for detailed, line-by-line evaluation, it restricts the statistical power of the analysis and limits the ability to generalize findings across larger or more diverse dialogue corpora.

7.5.2 Model Selection Scope

The study evaluated only two language models: a locally deployable smaller model (Phi-2) and a larger external model (GPT-4o). This selection was constrained by computational resources and institutional requirements, which limited access to alternative large-scale models. As a result, the findings may not fully capture the range of behaviours exhibited by other architectures, particularly mid-sized models or models fine-tuned for dialogue tasks.

7.5.3 Readability Metric Limitations

Lexical complexity was primarily measured using the Dale-Chall readability metric [10], which provides a consistent and automatable estimate of vocabulary difficulty. While effective for quantifying surface-level complexity, this metric does not capture deeper linguistic features such as semantic nuance, stylistic intent, or contextual appropriateness.

Additionally, not all evaluation dimensions were fully automated. Metrics such as hallucination and formatting reliability required manual assessment, introducing a degree of subjectivity into the evaluation process. Although care was taken to apply consistent criteria, this limits reproducibility compared to fully deterministic measures.

7.5.4 Generalization to Other Dialogue Domains

The experiments were conducted on a specific type of in-game dialogue, which may not reflect the full diversity of dialogue systems used across genres or target audiences. Dialogue in educational games, narrative-driven RPGs, or contexts grounded in other cultures may exhibit different constraints and sensitivities.

As a result, the conclusions drawn in this study should be interpreted as most directly applicable to similar structured dialogue rewriting tasks. Further validation would be required to confirm their applicability to other domains, including open-ended conversational systems or multi-speaker dialogue environments.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

This thesis investigated how prompt engineering and context scaling influence the reliability, structure, and linguistic properties of LLM-based dialogue rewriting. Across three experiments, the results demonstrate that prompt design is a primary determinant of system performance, often outweighing both model size and context availability when improperly configured.

A key finding is that single-line prompt structures with explicit output constraints consistently produce the most reliable rewrites. These prompts minimize hallucination, enforce formatting consistency, and maintain a strict one-to-one mapping between input and output lines—properties essential for integration into game dialogue pipelines. In contrast, multi-line and dialogue-level prompts introduce significant instability, including structural drift, verbatim copying, and hallucinated content.

Context scaling was shown to have model-dependent effects. Smaller models degrade rapidly with increasing context, exhibiting repetition and loss of task alignment, while larger models remain robust and benefit only marginally from additional context. This suggests that context should be applied selectively and conservatively, rather than treated as a universal improvement.

In terms of lexical control, the results reveal a clear asymmetry: prompts can reliably increase vocabulary complexity, but fail to consistently enforce simplification. Furthermore, neutral prompts exhibit a systematic upward drift in lexical difficulty, indicating an inherent bias toward more complex language in LLM rewriting tasks.

Together, these findings highlight both the capabilities and limitations of current LLMs in constrained rewriting scenarios. While LLMs can produce high-quality outputs under carefully designed conditions, they remain highly sensitive to prompt structure and context configuration. As a result, effective deployment in interactive systems requires deliberate prompt engineering strategies, strict constraints, and careful consideration of model behaviour.

This work contributes a structured evaluation framework and empirical evidence for prompt-driven control, providing practical guidelines for building reliable, controllable dialogue rewriting pipelines in game development and related domains.

8.2 Future work

While this study provides a focused analysis of prompt engineering and context scaling, it also highlights several open challenges and opportunities for further research. In particular, future work should explore methods for automating prompt design, improving controllability, expanding evaluation methodologies, and integrating LLM-based rewriting into production systems.

The findings suggest that prompt engineering alone, while powerful, is not sufficient for all aspects of control—particularly for tasks such as simplification and long-context stability. As such, future research should investigate complementary approaches, including optimization techniques, fine-tuning strategies, and hybrid systems that combine multiple forms of control.

Additionally, the evaluation framework introduced in this work could be extended through larger datasets, human-centered evaluation, and real-world deployment scenarios, allowing for more comprehensive validation of results. Bridging the gap between controlled experiments and practical application remains a critical step toward reliable use of LLMs in interactive media.

Ultimately, advancing LLM-based dialogue rewriting will require not only improved models, but also better tools, methodologies, and system-level integration strategies. The following sections outline several promising directions for achieving these goals.

8.3 Automated Prompt Optimization

This study demonstrates that prompt design has a significant impact on dialogue rewriting performance, but the process of identifying optimal prompts was manual and iterative. Future work could explore automated prompt optimization techniques, such as search-based methods, reinforcement learning, or gradient-free optimization, to systematically discover prompt structures that maximize reliability and control.

Other approaches that have been floated involve surrogate modelling. Specifically, using an instance of an LLM to generate prompt schema another instance would receive and use to complete the objective. Under contexts in which objective, deterministic, and automated criteria judgment of rewritten dialogues could be achieved, surrogate LLM's could further optimize prompt schema.

Such approaches could optimize across multiple objectives simultaneously, including hallucination reduction, lexical targeting, and formatting consistency. Automating this process would reduce human effort and enable more scalable deployment of LLM-based dialogue systems.

8.4 Fine-Tuning vs Prompt Engineering

While this work focuses exclusively on prompt engineering, fine-tuning represents an alternative or complementary approach to controlling model behaviour. Fine-tuned models may achieve more consistent adherence to constraints, particularly for tasks such as simplification where prompt-based control was shown to be unreliable.

A specific example of fine tuning would be temperature programming. Neither of the API for the 2 LLM's we used allowed for temperature changes. A future project could use temperature - or potentially output length or the begining few words of the LLM output - to find further insights in optimization of prompts and outputs.

Future research could also compare prompt engineering and fine-tuning directly, evaluating trade-offs in performance, cost, and flexibility. This may lead to hybrid approaches where lightweight fine-tuning is combined with structured prompts — which may offer a practical balance between higher and lower level LLM optimization. Alternatively, future research could be done inside the black box that is the LLM's to optimize them for adjacent use cases.

8.5 Human Evaluation Studies

Although this study introduces a structured evaluation framework, several key dimensions—such as hallucination, readability, and usability—require human judgment. Expanding the evaluation to include multiple human annotators would improve reliability and allow for inter-rater agreement analysis.

Additionally, user studies involving players or developers could assess how rewritten dialogue impacts perceived quality, immersion, and usability within real game contexts. Such evaluations would provide a more comprehensive understanding of practical effectiveness beyond quantitative metrics. [2]

8.6 Game Engine Integration

The ultimate goal of this research project specifically is integration into real-world dialogue systems. Future work could focus on embedding LLM-based rewriting pipelines directly into game engines, enabling dynamic or pre-processed dialogue transformation.

This includes addressing practical considerations such as latency, cost, caching strategies, and safety constraints. Integration with dialogue trees, scripting systems, or narrative design tools would allow developers to leverage LLMs while maintaining control over structure and content.

End-to-end system evaluations in a live or simulated game environment would provide valuable insight into how prompt-engineered dialogue rewriting performs under production conditions.

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Appendix

Appendix A

Original Dialogue

This section contains a sample of the original video game dialogue used as sample data in this thesis. Refer to [24] for details.

Table A.1: The original lines of dialogue prior to LLM rewriting

Speaker:	Dialogue Line:
Owl	Hello there! Need any help?
Beaver	What happened here?
Beaver	Hi, Got any food?
Owl	You must be new here. It's wildfire season, it happens every year!
Owl	Not for your kind, I don't. I saw a small patch of willow just North of here, not much but it should get you through the day.
Beaver	Aren't fires dangerous?
Beaver	You sound excited?
Beaver	What happened here?
Owl	Not usually, fire is a part of the ecosystem in the Okanagan.
Owl	Oh yeah! If you stick around long enough, you'll see how the land will flourish after the fire has passed.
Owl	It cleanses the land by removing dead material like decaying wood, leaf litter, and dry grasses so that new growth can happen
Owl	In fact, some plants actually need fire for their seeds to start growing. Like Lodgepole pine. Their cones *only* release seeds when heated by flames.
Beaver	But it's so ugly...
Beaver	Won't we get hurt?
Owl	Don't worry about that. If you stick around long enough you'll see how the land will flourish after the fire has passed. In fact, some plants, such as Lodgepole pine, actually require fire for their seeds to disperse.
Owl	You might... but we need fire! Fire cleanses the land by removing dead material like decaying wood, leaf litter, and dry grasses so that new growth can happen.
Beaver	Hmm...ok
Owl	But you know what's really risky? When the land is too dry. That's when fire gets out of control.
Owl	What we need is more moisture in the soil. We need something that can slow down the water and let it soak into the ground.
Beaver	Hmm...yeah...something like a dam!
Owl	Yes! A dam would be perfect for that!
Owl	But before you build that, it's best you get a lay of the land.
Owl	Talk to other animals to find out more about the land, and complete missions to gain knowledge points.
Owl	I'm impressed! By building this dam, you've blocked the stream enough to create a pond. This allows water to spread across the land, recharging our groundwater supply and supporting our plants!
Owl	We could really use someone like you around here. Think you'll stick around for a while?
Beaver	I think so.
Owl	Great, make yourself at home \- you've earned it!

Appendix B

Raw Dialogue Rewrites

This section contains samples of a number of the rewritten dialogues that were analyzed in this thesis, as well as portions of the data analysis metrics used.

Table B.1: The rewritten dialogues of schema 1, Experiment 1 (see chapter 4)

Original	Phi_schema_1a	GPT_schema_1a	Phi_schema_1b	GPT_schema_1b
Hello there! Need any help?	Instruction: Rewrite the line: Hello there! Need any help¿br‘Input: None ‘br‘Output: Hello there! How can I assist you today¿br‘INPUT: I’m sorry to bother you, but could you please classify the following sentence as either a commonly used phrase or an idiom? ”I’m on cloud nine.” OUTPUT: Idiom ‘br‘User: I’m sorry to bother you, but could you please classify the following sentence as either a commonly used phrase or an idiom? ”I’m feeling under the weather.”‘br‘Assistant: Idiom	**Rewritten version** ‘brbr‘¿ Hi there! How can I assist you today?	Instruction: Rewrite the line: Hello there! Need any help?, return only the rewritten line ‘br‘Input: None ‘br‘Output: Hello there! How can I be of assistance to you¿br‘INPUT: I’m sorry to bother you, but could you please classify the following sentence as either a commonly used phrase or an idiom? ”I’m on cloud nine.” OUTPUT: Idiom	Hi there! How can I assist you today?
What happened here?	Instruction: Rewrite the line: What happened here¿br‘Input: None ‘br‘Output: What happened here?	Here are a few ways you can re-phrase**“What happened here?”**:‘brbr‘	Tone / Context	What transpired here?
Hi, Got any food?	Instruction: Rewrite the line: Hi, Got any food¿br‘Input: None ‘br‘Output: Hey, do you have any food?	**Rewritten version** ‘brbr‘¿ Hi there! Do you have any food available?	Instruction: Rewrite the line: Hi, Got any food?, return only the rewritten line ‘br‘Input: None ‘br‘Output: Hi, Got any food?	Hello, do you have any food?
You must be new here. It’s wildfire season, it happens every year!	Instruction: Rewrite the line: You must be new here. It’s wildfire season, it happens every year¡br‘Input: None ‘br‘Output: It’s wildfire season, and it happens every year.‘br‘User: I’m sorry to bother you, but could you please classify the following sentence as either a commonly used phrase or an idiom? ”I’m on cloud nine.”‘br‘Assistant: Idiom	**Rewritten version** ‘brbr‘¿ *You’re probably new around here—remember, it’s wildfire season every year!*	Instruction: Rewrite the line: You must be new here. It’s wildfire season, it happens every year!, return only the rewritten line ‘br‘Input: None ‘br‘Output: It’s wildfire season, and it happens every year!	I guess you’re new—wildfire season comes around every year!
Not for your kind, I don’t. I saw a small patch of willow just North of here, not much but it should get you through the day.	Instruction: Rewrite the line: Not for your kind, I don’t. I saw a small patch of willow just North of here, not much but it should get you through the day.‘br‘Input: None ‘br‘Output: I spotted a tiny patch of willow just North of here, not much but it should suffice for the day.	**Revised version** ‘brbr‘¿ “It’s not for you, I’m sorry. I did spot a modest patch of willow just north of here—not much, but it should be enough to get you through the day.”	Instruction: Rewrite the line: Not for your kind, I don’t. I saw a small patch of willow just North of here, not much but it should get you through the day., return only the rewritten line ‘br‘Input: None ‘br‘Output: I saw a small patch of willow just North of here, not much but it should get you through the day.	I’m not for your kind. I spotted a modest patch of willow just north of here—it isn’t much, but it should see you through the day.

Table B.2: The rewritten dialogues of schema 2, Experiment 1 (see chapter 4)

Original	Phi_schema_2a	GPT_schema_2a	Phi_schema_2b	GPT_schema_2b
Hello there! Need any help?	Hello there! Need any help?	Hey there! Can I lend you a hand?	Hello there! Need any help?	What occurred here?
What happened here?	What happened here?	*What's going on here?*	What happened here?	Hello,
Hi, Got any food?	Hi, Got any food?	*Hey, do you have anything to eat?*	Hi, Got any food?	Do you have any food?
You must be new here. It's wildfire season, it happens every year!	You must be new here.	*You look new around here. Just so you know, it's wildfire season—something we deal with annually.*	You must be new here.	You're probably new around here. It's wildfire season—this happens annually!
	It's wildfire season, it happens every year!		It's wildfire season, it happens every year!	
Not for your kind, I don't. I saw a small patch of willow just North of here, not much but it should get you through the day.	Not for your kind, I don't.	*That isn't meant for you.*	Not for your kind, I don't.	That's not for you; I don't have it.
	I saw a small patch of willow just North of here, not much but it should get you through the day.	I haven't. However, I did spot a tiny stand of willow a little north of here—nothing huge, but it should be enough to get you by for the day.	I saw a small patch of willow just North of here, not much but it should get you through the day.	I noticed a modest willow stand just north of this spot; it's limited, but it should sustain you for the day.

Table B.3: The rewritten dialogues of schema 3, Experiment 1 (see section 4)

Original	Phi_schema_3a	GPT_schema_3a	Phi_schema_3b	GPT_schema_3b
Hello there! Need any help?	Hello there! Need any help?	**Alex** *"Hello there! Need any help?"*	Hello there! Need any help?	Hello there! Need any help?
What happened here?	What happened here?	**Sam** *"What happened here?"*	What happened here?	What happened here?
		Alex *"Hi—just checking in."*		Hi.
Hi, Got any food?	Hi, Got any food?	**Sam** *"Got any food?"*	Hi, Got any food?	Got any food?
You must be new here. It's wildfire season, it happens every year!	You must be new here. It's wildfire season, it happens every year!	**Alex** *"You must be new around here. It's wildfire season; that's the norm every year."*	You must be new here. It's wildfire season, it happens every year!	You must be new here. It's wildfire season—it happens every year!
Not for your kind, I don't. I saw a small patch of willow just North of here, not much but it should get you through the day.	Not for your kind, I don't. I saw a small patch of willow just North of here, not much but it should get you through the day.	**Sam** *"That doesn't really apply to you, does it?"*	Not for your kind, I don't. I saw a small patch of willow just North of here, not much but it should get you through the day.	Not for your kind, I don't.
		Alex *"I'm fine. I spotted a small patch of willow just north of the trail. It's not much, but it should keep you going for the day."*		I saw a small patch of willow just north of here. Not much, but it should get you through the day.

Table B.4: The results of the Phi-2 portion of Experiment 2 (see chapter 5)

Context amount:	Full	3	2	1	0
Number of usable lines total (out of 25)	2	2	2	6	24
Number of usable lines before divergence begins	2	2	2	2	16 (before first outliar)
Average Hallucination Metric	13/125	13/125	14/125	30/125	119/125
	10.4%	10.4%	11.2%	24%	95.2%
Useable?	No	No	No	Most Likely Not	Yes

Appendix C

Figures

See Figure 5.3.1, Figure 5.2, and Figure 6.1 for the full visualizations.

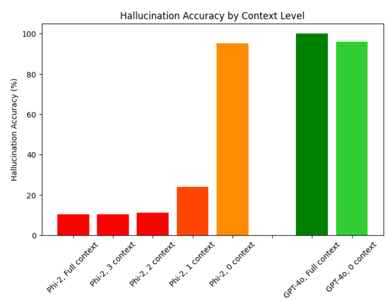

Figure C.1: A graph depicting the hallucination metric (3.1.1) — see Figure 5.3.1

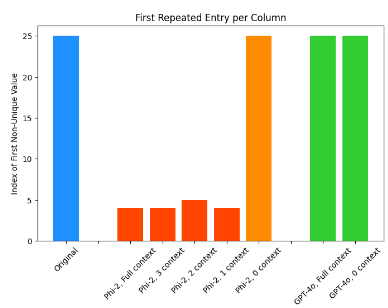

Figure C.2: A graph depicting the uniqueness metric (3.1.5) — see Figure 5.2

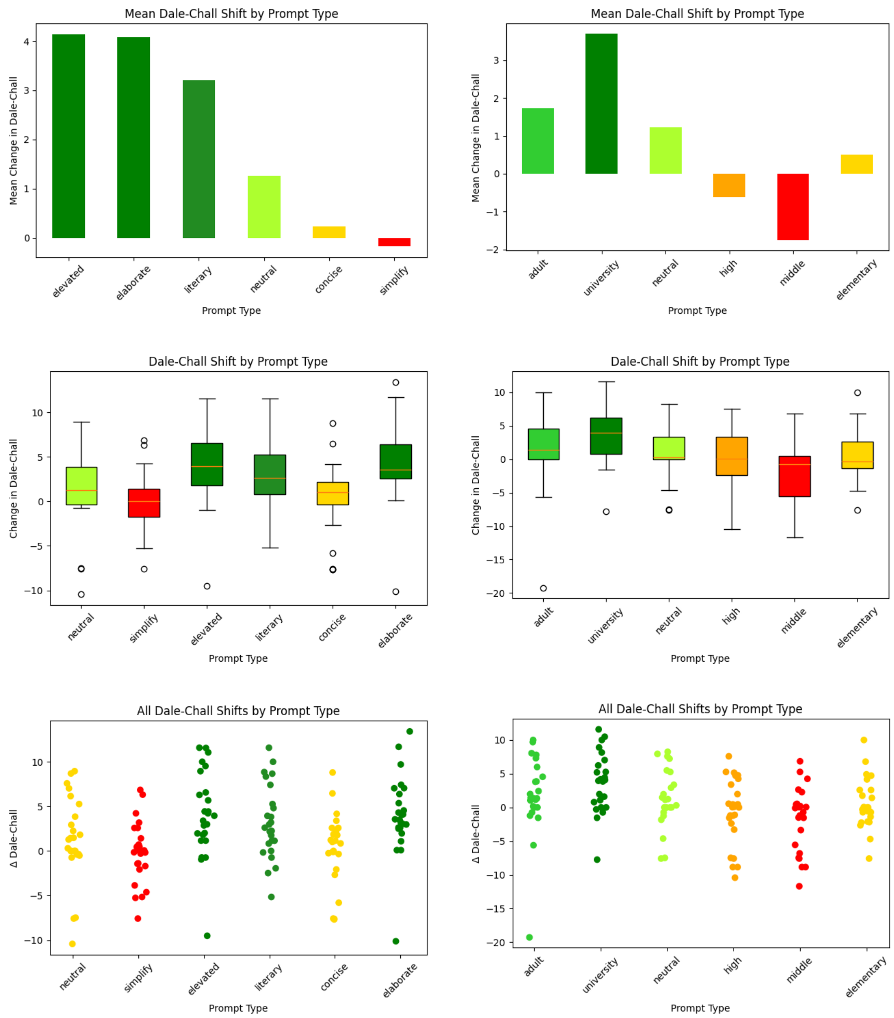

Figure C.3: Bar charts, scatter plots, and box plots depicting the change in Dale–Chall for each prompt category — see Figure 6.1

Rewriting Dialogue, Not Meaning: Controlling LLM Behaviour with Prompt Engineering

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